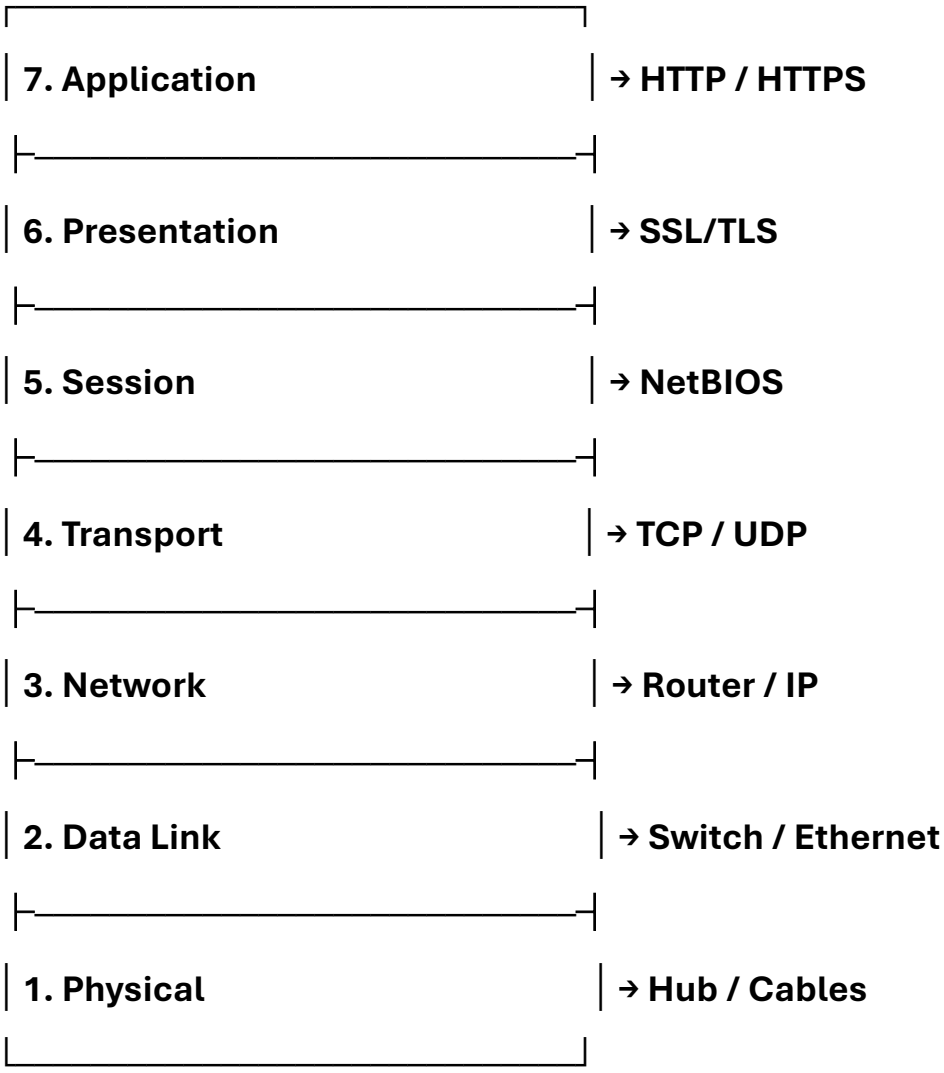
1. **Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application)**

OSI 7-LAYER MODEL



2.

Match the following networking scenarios to the correct OSI layer: a) Encrypting data before sending, b) Routing packets across networks, c) Converting data to electrical signals, d) Opening a web page in your browser, e) Detecting and correcting errors in frames.

Scenario	**OSI Layer**
a) Encrypting data before sending	Layer 6 – Presentation
b) Routing packets across networks	Layer 3 – Network
c) Converting data to electrical signals	Layer 1 – Physical
d) Opening a web page in your browser	Layer 7 – Application
e) Detecting and correcting errors in frames	Layer 2 – Data Link

3. **Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.**

➢ **Example: WhatsApp**

When I send a message on WhatsApp, different OSI layers work together:

1. **Application Layer (Layer 7):** WhatsApp sends the message using application protocols.
2. **Transport Layer (Layer 4):** TCP/UDP helps transport the data between my phone and WhatsApp's server.
3. **Network Layer (Layer 3):** IP addresses and routers move the data from my phone to the WhatsApp server.
4. **Data Link/Physical (Layers 2 & 1):** Wi-Fi or mobile network carries the data as signals.

▪ **So, the message travels through multiple OSI layers before reaching the receiver.**

4. **Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.**

➢ Possible OSI Layers: Layer 7 and Layer 4

If you can ping the website's IP address, the lower layers, especially Network Layer (Layer 3), are working.

The problem could be:

- **Layer 7 – Application: Website/server service may be down, or HTTP/HTTPS may have a problem.**
- **Layer 4 – Transport: TCP connection to ports 80 (HTTP) or 443 (HTTPS) may be blocked.**